



Andrea Campos Villarraga

Summary

Bilingual Energy Engineering Lead with extensive experience in residential solar PV project management, from feasibility and design to permitting and closeout. Proficient in Aurora Solar, AutoCAD, and Tesla Energy solutions. Proven ability to coordinate cross-functional teams, manage stakeholders, and ensure compliance with building codes and utility requirements. Seeking to leverage technical expertise and project lifecycle coordination skills to drive successful solar projects at a forward-thinking company.

Work Experience

2024 – Present	Energy Engineering Lead Bilingual Solar Engineer & Project Coordinator WINDMAR HOME FLORIDA - Provide technical leadership and project coordination for residential solar PV projects from feasibility and engineering through permitting, inspections, utility interconnection and PTO/closeout. - Oversee the full project lifecycle, including site assessment, energy analysis, system sizing, pre-engineering, design, permitting, inspections and final closeout. - Develop and review residential PV designs, engineering plans and technical documentation using Aurora Solar and AutoCAD. - Coordinate permit packages and technical documentation with Authorities Having Jurisdiction (AHJs), addressing permitting comments and technical requirements. - Orchestrate structural, electrical, and final inspections, maintaining communication among engineering, field teams, AHJs, utilities, and project stakeholders. - Review single-line diagrams, electrical layouts, site plans, and technical drawings for design quality and project/code compliance. - Facilitate utility interconnection and net-metering processes, applying applicable Florida Building Code and NEC requirements. - Develop hands-on expertise with Tesla Energy solutions, including Powerwall BESS and Wall Connector EV charging equipment; Tesla-certified for applicable products. - Analyze energy consumption and production data to inform system sizing, performance evaluation, and customer energy solutions.	Orlando, USA
05/2022 – 09/2023	Data Analyst i-APS - Managed data mapping, review, cleaning, visualization, and analysis for international humanitarian projects. - Supported Monitoring, Evaluation and Learning (MEL) and Third-Party Monitoring (TPM) activities for Colombia and international country operations.	Washington, USA

Contact

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Skills

Energy and Solar
Solar PV Project Management
Project Lifecycle Coordination
Residential PV Engineering and Design
Feasibility and Pre-Engineering
Energy Analysis and System Sizing
BESS
Off-Grid and Grid-Connected Systems
Solar Water Pumping
Energy Efficiency
Climate Change and Sustainability
Technical Project Coordination
Stakeholder Management
Permitting and AHJ Coordination
Inspection Coordination
Technical Documentation
Engineering Plan and Drawing Review
Project Reporting
Requirements and Deliverables Tracking
Aurora Solar
AutoCAD
PVsyst
RETScreen
Microsoft Project

- Authored and reviewed project reports and technical documentation in English and Spanish.
- Served as a focal point for climate change, sustainability, and renewable energy topics, bolstering internal knowledge and training initiatives.
- Participated as an internal auditor for ISO 9001 processes.

11/2021 – 01/2022

Junior Project Consultant

Bogota, Colombia

UNIVERSIDAD EAN / UNDP

- Assisted in the analysis of energy subsidies and contributed to project activities related to energy policy formulation.
- Executed research, analysis, and provided technical support for energy-sector projects.

08/2020 – 12/2021

Energy Access Volunteer Coordinator

Bogota, Colombia

FUNDACIÓN MGM

- Supported renewable energy initiatives addressing the energy needs of rural and vulnerable communities.
- Advised the implementation of a 3,750 Wp photovoltaic system in Quiba Guabal.
- Delivered renewable energy training to teachers and students through environmental education activities and the Development and Peace Program of Western Boyacá (BOYAPAZ).

08/2020 – 12/2021

Renewable Energy Project Coordinator

Bogota, Colombia

MGM INGENIERÍA Y PROYECTOS

- Coordinated photovoltaic and energy-efficiency projects, overseeing technical requirements, procurement, personnel, certifications, permits, and project execution.
- Supported the coordination of a photovoltaic installation for the Ministry of Mines and Energy, covering its main and archive facilities, with a nominal capacity of 100 kWAC.
- Continued to support renewable energy projects in non-interconnected and social-impact contexts, energy-efficiency initiatives, and off-grid systems.
- Contributed to projects involving Battery Energy Storage Systems (BESS), energy efficiency, and solar water pumping; performed system sizing and energy analysis using PVsyst and RETScreen.

2020 – 2020

Junior Sustainability Projects Consultant

Bogota

UNIVERSIDAD EAN / CAEM / UNDP

- Supported the NAMA Industry and Sustainability Plan focused on energy efficiency in the Bogotá–Cundinamarca industrial sector by collecting secondary information and statistical energy data.

2019 – 2019

Project Assistant

Bogota

DUCTECOL

Power BI

Google Data Studio

Microsoft Office

LTspice

Cross-Functional Communication

Bilingual Client and Stakeholder Support

Languages

Spanish (Native / Fluent)



English (Fluent / Bilingual professional proficiency)



- Assisted with thermal-load calculations, AutoCAD drawings, and quotations for HVAC, ventilation, and duct systems.

2016 – 2018 **Management Assistant** *Orange, USA*
POWER SERVICES

- Supported reporting, invoicing, scheduling, and coordination of technical maintenance personnel.

2012 – 2016 **Bilingual Customer Service Agent** *Ohio, USA*
AT&T; U-verse

- Provided bilingual customer service, billing, sales, and technical support for a U.S. telecommunications operation.

Education

01/2021 – 10/2021 **Universidad EAN** *Bogota*
Specialist in Project Management

2017 – 05/2021 **Universidad EAN** *Bogota*
Energy Engineer, Energy Engineering

- Graduated May 2021.

Certificates

ISO 50001 Energy Management Systems Auditor
ISO 9001 Internal Auditor

Tesla Energy Certification

Energy Leadership Training Program

Climate Action Education

Sustainable Project Trends for Industry